Solutions to TD 8 & 9: Risk Measures

ST7 Stochastic Finance and Risk Modelling

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1 VaR and ES for common distributions

(1) $L \sim \mathbf{Exp}(\lambda)$

The probability density function is $f(x) = \lambda e^{-\lambda x}$ and the cumulative distribution function (CDF) is $F_L(x) = \int_0^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}$ for $x \geq 0$.

To find the Value at Risk (VaR_α), we use the definition and solve for $F_L(x) = \alpha$:

$$\begin{aligned} 1 - e^{-\lambda x} &= \alpha \\ e^{-\lambda x} &= 1 - \alpha \\ -\lambda x &= \ln(1-\alpha) \\ VaR_\alpha(L) &= -\frac{1}{\lambda}\ln(1-\alpha) \end{aligned}$$

Since the exponential distribution is continuous, the Expected Shortfall (ES_α) can be computed using the conditional expectation $ES_\alpha(L) = \mathbb{E}[L|L \geq VaR_\alpha(L)]$. Thanks to the memoryless property of the exponential distribution, the excess loss over a threshold x, given that it exceeds x, has the exact same Exp(λ) distribution. Therefore:

$$\begin{aligned} ES_\alpha(L) &= VaR_\alpha(L) + \mathbb{E}[L] \\ ES_\alpha(L) &= -\frac{1}{\lambda}\ln(1-\alpha) + \frac{1}{\lambda} \end{aligned}$$

(2) $L \sim \mathbf{Unif}([a,b])$

The CDF for a uniform distribution on $[a,b]$ is $F_L(x) = \frac{x-a}{b-a}$ for $x \in [a,b]$.

To find $VaR_\alpha(L)$, we set the CDF equal to α:

$$\begin{aligned} \frac{x-a}{b-a} &= \alpha \\ x - a &= \alpha(b-a) \\ VaR_\alpha(L) &= a + \alpha(b-a) = (1-\alpha)a + \alpha b \end{aligned}$$

Since the distribution is continuous, $ES_\alpha(L) = \mathbb{E}[L|L \geq VaR_\alpha(L)]$. Conditioned on being greater than or equal to $VaR_\alpha(L)$, the random variable L follows a new uniform distribution on the interval $[VaR_\alpha(L), b]$. The expected value of a uniform distribution is simply its midpoint:

$$\begin{aligned} ES_\alpha(L) &= \frac{VaR_\alpha(L) + b}{2} \\ ES_\alpha(L) &= \frac{a + \alpha(b-a) + b}{2} \end{aligned}$$

(3) $L \sim \chi^2(1)$ i.e., $L = G^2$ with $G \sim \mathcal{N}(0,1)$

First, let's find the CDF of L. Let Φ be the CDF of the standard normal distribution.

$$\begin{aligned} F_L(x) &= \mathbb{P}(G^2 \leq x) \\ &= \mathbb{P}(-\sqrt{x} \leq G \leq \sqrt{x}) \\ &= \Phi(\sqrt{x}) - \Phi(-\sqrt{x}) \end{aligned}$$

Because the standard normal is symmetric, $\Phi(-\sqrt{x}) = 1 - \Phi(\sqrt{x})$. Thus:

$$F_L(x) = 2\Phi(\sqrt{x}) - 1$$

To find $VaR_\alpha(L)$, we solve $F_L(x) = \alpha$:

$$\begin{aligned} 2\Phi(\sqrt{x}) - 1 &= \alpha \\ \Phi(\sqrt{x}) &= \frac{1+\alpha}{2} \\ \sqrt{x} &= \Phi^{-1}\left(\frac{1+\alpha}{2}\right) \\ VaR_\alpha(L) &= \left[\Phi^{-1}\left(\frac{1+\alpha}{2}\right)\right]^2 \end{aligned}$$

For the Expected Shortfall, we use the integral definition. Let $c = \sqrt{VaR_\alpha(L)} = \Phi^{-1}\left(\frac{1+\alpha}{2}\right)$.

$$ES_\alpha(L) = \mathbb{E}[G^2 | G^2 \geq VaR_\alpha(L)] = \frac{1}{1-\alpha}\mathbb{E}[G^2 1_{\{|G| \geq c\}}]$$

By symmetry of the standard normal density $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$:

$$\mathbb{E}[G^2 1_{\{|G| \geq c\}}] = 2\int_c^\infty z^2 \phi(z) dz$$

We evaluate this using integration by parts, setting $u = z$ and $dv = z\phi(z)dz$ (which implies $v = -\phi(z)$):

$$\begin{aligned} \int_c^\infty z^2\phi(z)dz &= [-z\phi(z)]_c^\infty + \int_c^\infty \phi(z)dz \\ &= c\phi(c) + (1 - \Phi(c)) \end{aligned}$$

Notice that $\Phi(c) = \frac{1+\alpha}{2}$, which means $1 - \Phi(c) = \frac{1-\alpha}{2}$. Therefore:

$$2\int_c^\infty z^2\phi(z)dz = 2c\phi(c) + 2\left(\frac{1-\alpha}{2}\right) = 2c\phi(c) + 1 - \alpha$$

Finally, plugging this back into the Expected Shortfall equation:

$$\begin{aligned} ES_\alpha(L) &= \frac{2c\phi(c) + 1 - \alpha}{1-\alpha} \\ ES_\alpha(L) &= 1 + \frac{2c\phi(c)}{1-\alpha} \quad \text{where} \quad c = \Phi^{-1}\left(\frac{1+\alpha}{2}\right) \end{aligned}$$

(4) L has the discrete distribution

From the table provided, the possible loss values l_i and their probabilities $p_i = \mathbb{P}(L = l_i)$ are:

l_i	-5	0	3	6	8	10
p_i	0.10	0.50	0.23	0.07	0.06	0.04
$F_L(l_i)$	0.10	0.60	0.83	0.90	0.96	1.00

Computing $VaR_{0.95}(L)$: Recall that $VaR_\alpha(L) = \inf\{x \in \mathbb{R} : F_L(x) \geq \alpha\}$. Looking at the cumulative probabilities, the smallest value x for which $F_L(x) \geq 0.95$ is $x = 8$ (since $F_L(8) = 0.96$).

$$VaR_{0.95}(L) = 8$$

Computing $VaR_{0.99}(L)$: Similarly, the smallest value x for which $F_L(x) \geq 0.99$ is $x = 10$ (since $F_L(10) = 1.00$).

$$VaR_{0.99}(L) = 10$$

Computing $ES_{0.95}(L)$: For a discrete distribution, we must use the integral definition: $ES_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 VaR_u(L)du$. We can break the integral into intervals where VaR_u is constant: - For $u \in [0.95, 0.96)$, $VaR_u(L) = 8$. - For $u \in [0.96, 1.00]$, $VaR_u(L) = 10$.

$$\begin{aligned}
ES_{0.95}(L) &= \frac{1}{1-0.95}\left(\int_{0.95}^{0.96} 8\,du + \int_{0.96}^{1.00} 10\,du\right) \\
&= \frac{1}{0.05}\left(8(0.01) + 10(0.04)\right) \\
&= 20\left(0.08 + 0.40\right) \\
&= 20(0.48) = 9.6
\end{aligned}$$

Computing $ES_{0.99}(L)$: For $u \in [0.99, 1]$, $VaR_u(L) = 10$.

$$\begin{aligned}
ES_{0.99}(L) &= \frac{1}{1-0.99}\int_{0.99}^{1} 10\,du \\
&= \frac{1}{0.01}(10 \times 0.01) \\
&= 10
\end{aligned}$$

2 Monotone transformations

(1) Continuous and strictly increasing transformations

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous and strictly increasing function. We want to show that $VaR_\alpha(f(L)) = f(VaR_\alpha(L))$.

First, recall the definition of Value at Risk:

$$VaR_\alpha(L) = \inf\{x \in \mathbb{R} : \mathbb{P}(L \leq x) \geq \alpha\}$$

Now, let us find the cumulative distribution function for $f(L)$. Since f is strictly increasing, it is invertible, and the event $\{f(L) \leq x\}$ is equivalent to the event $\{L \leq f^{-1}(x)\}$.

$$\mathbb{P}(f(L) \leq x) = \mathbb{P}(L \leq f^{-1}(x)) = F_L(f^{-1}(x))$$

Applying the definition of VaR to $f(L)$:

$$VaR_\alpha(f(L)) = \inf\{x \in \mathbb{R} : \mathbb{P}(f(L) \leq x) \geq \alpha\}$$

$$VaR_\alpha(f(L)) = \inf\{x \in \mathbb{R} : F_L(f^{-1}(x)) \geq \alpha\}$$

Let us perform a change of variable by setting $y = f^{-1}(x)$, which implies $x = f(y)$. Because f is strictly increasing, as x increases, y also increases. Furthermore, taking the infimum over x is equivalent to taking the infimum over y and then applying f, due to the fact that f is continuous and preserves the ordering of the set.

$$\begin{aligned} VaR_\alpha(f(L)) &= \inf\{f(y) \in \mathbb{R} : F_L(y) \geq \alpha\} \\ &= f\left(\inf\{y \in \mathbb{R} : F_L(y) \geq \alpha\}\right) \\ &= f(VaR_\alpha(L)) \end{aligned}$$

(2) Continuous and strictly decreasing transformations

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous and strictly decreasing function. We want to show that $VaR_\alpha(f(L)) = f(VaR^+_{1-\alpha}(L))$.

Since f is strictly decreasing, the inequality direction reverses when applying the inverse function. The event $\{f(L) \leq x\}$ is equivalent to $\{L \geq f^{-1}(x)\}$.

$$\mathbb{P}(f(L) \leq x) = \mathbb{P}(L \geq f^{-1}(x))$$

We write the VaR for $f(L)$:

$$VaR_\alpha(f(L)) = \inf\{x \in \mathbb{R} : \mathbb{P}(L \geq f^{-1}(x)) \geq \alpha\}$$

Let $y = f^{-1}(x)$. Since f is strictly decreasing, as x decreases toward its infimum, y increases toward its supremum. Therefore, finding the infimum of x is equivalent to finding the supremum of y and applying f:

$$VaR_\alpha(f(L)) = f\left(\sup\{y \in \mathbb{R} : \mathbb{P}(L \geq y) \geq \alpha\}\right)$$

We must now show that $\sup\{y \in \mathbb{R} : \mathbb{P}(L \geq y) \geq \alpha\} = VaR^+_{1-\alpha}(L)$. Recall the definition of $VaR^+_{1-\alpha}(L)$:

$$VaR^+_{1-\alpha}(L) = \inf\{z \in \mathbb{R} : \mathbb{P}(L \leq z) > 1 - \alpha\}$$

Let $A = \{y \in \mathbb{R} : \mathbb{P}(L \geq y) \geq \alpha\}$ and $B = \{z \in \mathbb{R} : \mathbb{P}(L \leq z) > 1 - \alpha\}$. If $y \in A$, then $\mathbb{P}(L \geq y) \geq \alpha$. Since $\mathbb{P}(L \geq y) = 1 - \mathbb{P}(L < y)$, we have:

$$\begin{aligned} 1 - \mathbb{P}(L < y) &\geq \alpha \\ \mathbb{P}(L < y) &\leq 1 - \alpha \end{aligned}$$

For any $z < y$, it is strictly true that $\mathbb{P}(L \leq z) \leq \mathbb{P}(L < y) \leq 1 - \alpha$. This implies $z \notin B$. Therefore, any element in B must be greater than or equal to y. This means $y \leq \inf B$, which implies $\sup A \leq \inf B$.

Conversely, take any $y < \inf B$. By definition of the infimum of B, $y \notin B$, which means $\mathbb{P}(L \leq y) \leq 1-\alpha$. Consequently, $\mathbb{P}(L > y) = 1 - \mathbb{P}(L \leq y) \geq \alpha$. Since $\mathbb{P}(L \geq y) \geq \mathbb{P}(L > y) \geq \alpha$, it follows that $y \in A$. Since every value less than $\inf B$ is in A, we must have $\sup A \geq \inf B$.

Combining $\sup A \leq \inf B$ and $\sup A \geq \inf B$, we conclude $\sup A = \inf B$. Thus:

$$\sup\{y \in \mathbb{R} : \mathbb{P}(L \geq y) \geq \alpha\} = \inf\{z \in \mathbb{R} : \mathbb{P}(L \leq z) > 1 - \alpha\} = VaR^{+}_{1-\alpha}(L)$$

Substitute this back into the equation for $VaR_\alpha(f(L))$:

$$VaR_\alpha(f(L)) = f(VaR^{+}_{1-\alpha}(L))$$

Deduction for continuous F_L:

We are given the property that if F_L is continuous, $VaR_\beta(L) = VaR^{+}_{\beta}(L)$ for any confidence level $\beta \in (0, 1)$. Applying this property to $\beta = 1 - \alpha$, we get:

$$VaR^{+}_{1-\alpha}(L) = VaR_{1-\alpha}(L)$$

Substituting this into the result we just proved for strictly decreasing functions:

$$VaR_\alpha(f(L)) = f(VaR_{1-\alpha}(L))$$

3 The conditional definition of ES may fail

(1) Compute $VaR_{0.95}(L)$

The cumulative distribution function (CDF) of L is given by:

$$F_L(x) = \begin{cases} 0 & \text{if } x < 0 \\ 0.95 & \text{if } 0 \leq x < 100 \\ 1 & \text{if } x \geq 100 \end{cases}$$

By definition, $VaR_\alpha(L) = \inf\{x \in \mathbb{R} : F_L(x) \geq \alpha\}$. For $\alpha = 0.95$, the smallest value x such that $F_L(x) \geq 0.95$ is $x = 0$. Therefore,

$$VaR_{0.95}(L) = 0$$

(2) Compute $\mathbb{E}[L|L \geq VaR_{0.95}(L)]$

Since we found $VaR_{0.95}(L) = 0$, we need to compute $\mathbb{E}[L|L \geq 0]$.

Because the only possible values for L are 0 and 100, the condition $L \geq 0$ is always satisfied (it holds with probability 1). Thus, the conditional expectation is simply the unconditional expectation:

$$\begin{aligned} \mathbb{E}[L|L \geq 0] &= \mathbb{E}[L] \\ &= 0 \times \mathbb{P}(L = 0) + 100 \times \mathbb{P}(L = 100) \\ &= 0 \times 0.95 + 100 \times 0.05 \\ &= 5 \end{aligned}$$

(3) Compute $ES_{0.95}(L)$

The Expected Shortfall is defined via the integral:

$$ES_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 VaR_u(L) du$$

For $u \in (0.95, 1]$, we are looking for $\inf\{x : F_L(x) \geq u\}$. Since $u > 0.95$, we must have $F_L(x) = 1$, which first occurs at $x = 100$. Thus, $VaR_u(L) = 100$ for all $u \in (0.95, 1]$.

Now we compute the integral for $\alpha = 0.95$:

$$\begin{aligned} ES_{0.95}(L) &= \frac{1}{1-0.95} \int_{0.95}^1 100\, du \\ &= \frac{1}{0.05} \left[100u\right]_{0.95}^1 \\ &= 20 \times 100(1 - 0.95) \\ &= 20 \times 5 \\ &= 100 \end{aligned}$$

(4) Explain why the conditional expectation definition is problematic here

In this example, the conditional expectation yields 5, whereas the true Expected Shortfall yields 100. This shows a severe discrepancy.

The formula $ES_\alpha(L) = \mathbb{E}[L|L \geq VaR_\alpha(L)]$ is only guaranteed to hold when the cumulative distribution function F_L is continuous at $VaR_\alpha(L)$. Here, the distribution is discrete, and there is a massive probability mass exactly at $VaR_{0.95}(L) = 0$, since $\mathbb{P}(L = 0) = 0.95$.

When we condition on the event $L \geq VaR_{0.95}(L)$, we are inadvertently including the entire probability mass of the non-tail event $L = 0$. This "dilutes" the expectation with a large number of zero-loss scenarios, fundamentally misrepresenting the average of the strictly worst 5% of cases. The true integral definition of ES correctly isolates the 5% tail, which in this distribution consists entirely of the $L = 100$ outcome.

4 A regulator's paradox

4.1 (1) Compute VaR_0.95 of both strategies

For Strategy A, the cumulative distribution function $F_{L_A}(x)$ is:

$$F_{L_A}(x) = \begin{cases} 0 & \text{if } x < -1 \\ 0.95 & \text{if } -1 \leq x < 100 \\ 1 & \text{if } x \geq 100 \end{cases}$$

The Value at Risk is $VaR_{0.95}(L_A) = \inf\{x : F_{L_A}(x) \geq 0.95\}$. The smallest x satisfying this is -1.

$$VaR_{0.95}(L_A) = -1$$

For Strategy B, L_B is deterministic. The CDF jumps to 1 at $x = 0$.

$$VaR_{0.95}(L_B) = 0$$

4.2 (2) Which strategy satisfies regulation?

The regulatory constraint is $VaR_{0.95}(L) \leq 0$. Since $VaR_{0.95}(L_A) = -1 \leq 0$ and $VaR_{0.95}(L_B) = 0 \leq 0$, <u>both</u> strategies satisfy the regulation.

4.3 (3) Compare expected losses

Let us compute the expected loss for each strategy:

$$\begin{aligned} \mathbb{E}[L_A] &= (-1) \times \mathbb{P}(L_A = -1) + (100) \times \mathbb{P}(L_A = 100) \\ &= -1(0.95) + 100(0.05) \\ &= -0.95 + 5 \\ &= 4.05 \end{aligned}$$

For Strategy B, since the loss is always 0:

$$\mathbb{E}[L_B] = 0$$

Strategy A has a significantly higher expected loss (4.05) compared to Strategy B (0), despite both passing the VaR constraint.

4.4 (4) Explain why VaR may create incentives to take catastrophic risks

Value at Risk only looks at the probability of a loss, completely ignoring the severity of losses that occur beyond the chosen confidence level. By taking on a strategy like A, a bank can generate small, highly probable profits (the negative loss of -1) to satisfy the regulator, while hiding a catastrophic, low-probability risk (the loss of 100) in the tail. VaR incentivizes this behavior because it does not penalize the magnitude of extreme tail events.

4.5 (5) Show that ES removes this paradox

Let us compute the Expected Shortfall ($ES_{0.95}$) for both strategies.

For Strategy B, the loss is constant at 0, so $ES_{0.95}(L_B) = 0$.

For Strategy A, the quantile function $VaR_u(L_A) = 100$ for $u \in (0.95, 1]$.

$$\begin{aligned} ES_{0.95}(L_A) &= \frac{1}{1-0.95}\int_{0.95}^{1} VaR_u(L_A)du \\ &= \frac{1}{0.05}\int_{0.95}^{1} 100\,du \\ &= \frac{1}{0.05} \times 100(1-0.95) \\ &= 100 \end{aligned}$$

Unlike VaR, Expected Shortfall accounts for the magnitude of tail losses. Under an ES-based regulatory constraint (e.g., $ES_{0.95}(L) \leq 0$), Strategy A would fail spectacularly ($100 \not\leq 0$), successfully resolving the paradox.

4.6 (6) Construct a sequence L_n where expected loss diverges

We want to construct a sequence (L_n) such that $VaR_{0.95}(L_n) = 0$ for all n, but $\mathbb{E}[L_n] \to +\infty$.

Define the discrete random variable L_n as follows:

$$\begin{aligned} \mathbb{P}(L_n = 0) &= 0.95 \\ \mathbb{P}(L_n = n) &= 0.05 \end{aligned}$$

where $n \in \mathbb{N}, n > 0$.

For any n, $F_{L_n}(0) = 0.95$, which implies $VaR_{0.95}(L_n) = 0$. This perfectly satisfies the constraint. Now, we look at the expected loss:

$$\mathbb{E}[L_n] = 0(0.95) + n(0.05) = 0.05n$$

As $n \to +\infty$, the limit of the expected loss $\mathbb{E}[L_n] \to +\infty$.

4.7 (7) Interpret financially

This demonstrates that a financial institution can theoretically take on infinitely large catastrophic risks (the state $L_n = n$) without violating a VaR-based regulatory framework, provided the probability of that catastrophe is kept just at or below the regulatory threshold (here, 5%). The regulation fails to protect against the actual severity of systemic collapse.

4.8 (8) Compute ES_0.95(L_n) & comment

Let us calculate the Expected Shortfall for this sequence: For $u \in (0.95, 1]$, $VaR_u(L_n) = n$.

$$\begin{aligned} ES_{0.95}(L_n) &= \frac{1}{0.05}\int_{0.95}^{1} n\,du \\ &= \frac{1}{0.05} \times n(0.05) \\ &= n \end{aligned}$$

As $n \to +\infty$, $ES_{0.95}(L_n) \to +\infty$. Comment: Expected Shortfall accurately reflects the increasing tail risk. While VaR remains entirely blind to the growing catastrophic exposure, ES correctly identifies it and diverges alongside the potential magnitude of the loss.

5 Gaussian portfolio VaR: role of correlation

(1) Compute μ_P and σ_P^2 as functions of ρ

The portfolio return is a linear combination of the individual asset returns:

$$R_P = w_A R_A + w_B R_B$$

where $w_A = 0.35$ and $w_B = 0.65$.

Since the expectation is a linear operator, the expected portfolio return μ_P is independent of the correlation ρ:

$$\begin{aligned}
\mu_P &= \mathbb{E}[R_P] = w_A \mathbb{E}[R_A] + w_B \mathbb{E}[R_B] \\
&= w_A \mu_A + w_B \mu_B \\
&= 0.35(0.20) + 0.65(0.18) \\
&= 0.070 + 0.117 \\
&= 0.187
\end{aligned}$$

The variance of the portfolio return σ_P^2, however, depends on the covariance (and thus the correlation) between the two assets:

$$\begin{aligned}
\sigma_P^2(\rho) &= \mathrm{Var}(w_A R_A + w_B R_B) \\
&= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \mathrm{Cov}(R_A, R_B) \\
&= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho \sigma_A \sigma_B
\end{aligned}$$

Let us compute the individual terms:

- $w_A^2 \sigma_A^2 = (0.35)^2 (0.15)^2 = 0.1225 \times 0.0225 = 0.00275625$
- $w_B^2 \sigma_B^2 = (0.65)^2 (0.12)^2 = 0.4225 \times 0.0144 = 0.006084$
- $2 w_A w_B \sigma_A \sigma_B = 2(0.35)(0.65)(0.15)(0.12) = 0.00819$

Substituting these values back into the equation:

$$\begin{aligned}
\sigma_P^2(\rho) &= 0.00275625 + 0.006084 + 0.00819\rho \\
&= 0.00884025 + 0.00819\rho
\end{aligned}$$

(2) Deduce an analytical formula for $VaR_\alpha(L)$

The loss is defined as $L = -V_0 R_P = -100 R_P$. Since R_A and R_B are Gaussian, their linear combination R_P is also Gaussian. Thus, $R_P \sim \mathcal{N}(\mu_P, \sigma_P^2)$.

Because L is a simple scaling of R_P, the loss variable L is also normally distributed:

$$\begin{aligned}
\mu_L &= \mathbb{E}[-100 R_P] = -100 \mu_P \\
\sigma_L^2 &= \mathrm{Var}(-100 R_P) = (-100)^2 \sigma_P^2 = 10000 \sigma_P^2 \\
\sigma_L &= 100 \sigma_P
\end{aligned}$$

Therefore, $L \sim \mathcal{N}(-100\mu_P, 10000\sigma_P^2)$.

For any normally distributed random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, the Value at Risk at confidence level α is given by $VaR_\alpha(X) = \mu + \sigma \Phi^{-1}(\alpha)$, where Φ^{-1} is the inverse of the standard normal CDF.

Applying this to our loss L:

$$\begin{aligned}
VaR_\alpha(L) &= \mu_L + \sigma_L \Phi^{-1}(\alpha) \\
&= -100\mu_P + 100\sigma_P(\rho)\Phi^{-1}(\alpha)
\end{aligned}$$

(3) Compute $VaR_{0.99}(L)$ for $\rho = 0.5$ and $\rho = -0.5$. Comment.

First, we need the standard normal quantile for $\alpha = 0.99$. $\Phi^{-1}(0.99) \approx 2.3263$.

Recall $\mu_P = 0.187$, so $-100\mu_P = -18.7$.

Case 1: $\rho = 0.5$

We first compute the portfolio variance and standard deviation:

$$\begin{aligned}\sigma_P^2(0.5) &= 0.00884025 + 0.00819(0.5)\\ &= 0.00884025 + 0.004095\\ &= 0.01293525\end{aligned}$$

$$\sigma_P(0.5) = \sqrt{0.01293525} \approx 0.113733$$

Now we compute the VaR:

$$\begin{aligned}VaR_{0.99}(L) &= -18.7 + 100(0.113733)(2.3263)\\ &= -18.7 + 11.3733(2.3263)\\ &= -18.7 + 26.4577\\ &\approx 7.758 \text{ M euros}\end{aligned}$$

Case 2: $\rho = -0.5$

We compute the portfolio variance and standard deviation:

$$\begin{aligned}\sigma_P^2(-0.5) &= 0.00884025 + 0.00819(-0.5)\\ &= 0.00884025 - 0.004095\\ &= 0.00474525\end{aligned}$$

$$\sigma_P(-0.5) = \sqrt{0.00474525} \approx 0.068886$$

Now we compute the VaR:

$$\begin{aligned}VaR_{0.99}(L) &= -18.7 + 100(0.068886)(2.3263)\\ &= -18.7 + 6.8886(2.3263)\\ &= -18.7 + 16.0250\\ &\approx -2.675 \text{ M euros}\end{aligned}$$

Comment: The correlation coefficient strongly impacts the portfolio risk. When the assets are positively correlated ($\rho = 0.5$), they tend to move in the same direction, leading to a higher portfolio variance and a substantial $VaR_{0.99}$ of approximately 7.76 M euros. Conversely, when the assets are negatively correlated ($\rho = -0.5$), their price movements tend to offset each other, demonstrating the power of diversification. This drastically reduces the portfolio variance, pulling the $VaR_{0.99}$ down to approximately -2.68 M euros, meaning that even in the worst 1% of cases, the portfolio is actually expected to make a profit.

6 ES is coherent

(1) Show that ES_α is monotone and cash-invariant

Monotonicity: Let L_1 and L_2 be two loss random variables such that $L_1 \leq L_2$ almost surely. By the definition of Value at Risk, for any probability level $u \in (0,1)$, $VaR_u(L_1) \leq VaR_u(L_2)$. Using the integral representation of Expected Shortfall (which is directly equivalent to the conditional expectation for continuous distributions):

$$ES_\alpha(L_1) = \frac{1}{1-\alpha}\int_\alpha^1 VaR_u(L_1)du \leq \frac{1}{1-\alpha}\int_\alpha^1 VaR_u(L_2)du = ES_\alpha(L_2)$$

Thus, ES_α is monotone.

Cash-invariance (Translation invariance): Let $c \in \mathbb{R}$ be a deterministic constant (e.g., adding cash to a portfolio reduces the loss by c). We evaluate $ES_\alpha(L+c)$. First, note that $VaR_\alpha(L+c) = VaR_\alpha(L)+c$. Using the conditional expectation formula provided:

$$\begin{aligned}
ES_\alpha(L+c) &= \mathbb{E}[L+c \mid L+c \geq VaR_\alpha(L+c)] \\
&= \mathbb{E}[L+c \mid L+c \geq VaR_\alpha(L)+c] \\
&= \mathbb{E}[L+c \mid L \geq VaR_\alpha(L)] \\
&= \mathbb{E}[L \mid L \geq VaR_\alpha(L)] + c \\
&= ES_\alpha(L) + c
\end{aligned}$$

Thus, ES_α is cash-invariant.

(2) Show that ES_α is positively homogeneous

Let $\lambda > 0$ be a constant. We want to show $ES_\alpha(\lambda L) = \lambda ES_\alpha(L)$. Since $\lambda > 0$, the function $f(x) = \lambda x$ is strictly increasing. From the results of Exercise 2, we know that $VaR_\alpha(\lambda L) = \lambda VaR_\alpha(L)$. Substitute this into the provided formula:

$$\begin{aligned}
ES_\alpha(\lambda L) &= \mathbb{E}[\lambda L \mid \lambda L \geq VaR_\alpha(\lambda L)] \\
&= \mathbb{E}[\lambda L \mid \lambda L \geq \lambda VaR_\alpha(L)] \\
&= \mathbb{E}[\lambda L \mid L \geq VaR_\alpha(L)] \quad \text{(since } \lambda > 0\text{)} \\
&= \lambda \mathbb{E}[L \mid L \geq VaR_\alpha(L)] \\
&= \lambda ES_\alpha(L)
\end{aligned}$$

Thus, ES_α is positively homogeneous.

(3) Prove the subadditivity of ES_α

To prove $ES_\alpha(L_1+L_2) \leq ES_\alpha(L_1)+ES_\alpha(L_2)$, we first establish a useful lemma using the secondary formula given in the exercise:

$$ES_\alpha(L) = \frac{1}{1-\alpha}\mathbb{E}[L \cdot 1_{\{L \geq VaR_\alpha(L)\}}]$$

Lemma: For any event A such that $\mathbb{P}(A) = 1-\alpha$, we have $\mathbb{E}[L \cdot 1_A] \leq \mathbb{E}[L \cdot 1_{\{L \geq q\}}]$ where $q = VaR_\alpha(L)$.
Proof of Lemma: Let us look at the difference:

$$\mathbb{E}[L \cdot 1_{\{L \geq q\}}] - \mathbb{E}[L \cdot 1_A] = \mathbb{E}[L(1_{\{L \geq q\}} - 1_A)]$$

Consider the value of $L(1_{\{L \geq q\}} - 1_A)$ inside the expectation:

- If $L \geq q$, then $1_{\{L \geq q\}} = 1$. The term $(1_{\{L \geq q\}} - 1_A) \geq 0$. Since $L \geq q$, we have $L(1_{\{L \geq q\}} - 1_A) \geq q(1_{\{L \geq q\}} - 1_A)$.

- If $L < q$, then $1_{\{L \geq q\}} = 0$. The term $(1_{\{L \geq q\}} - 1_A) \leq 0$. Since $L < q$ and the term is negative or zero, we also have $L(1_{\{L \geq q\}} - 1_A) \geq q(1_{\{L \geq q\}} - 1_A)$.

In both cases, $L(1_{\{L \geq q\}} - 1_A) \geq q(1_{\{L \geq q\}} - 1_A)$ almost surely. Taking the expectation on both sides:

$$\begin{aligned}
\mathbb{E}[L(1_{\{L \geq q\}} - 1_A)] &\geq q\mathbb{E}[1_{\{L \geq q\}} - 1_A] \\
&= q(\mathbb{P}(L \geq q) - \mathbb{P}(A)) \\
&= q((1-\alpha) - (1-\alpha)) = 0
\end{aligned}$$

This proves the lemma. It demonstrates that $ES_\alpha(L)$ can be viewed as the supremum of $\frac{1}{1-\alpha}\mathbb{E}[L \cdot 1_A]$ over all possible events A with probability $1-\alpha$.

Proof of Subadditivity: Let $L_P = L_1 + L_2$ and define the tail event for the combined portfolio $A^* = \{L_P \geq VaR_\alpha(L_P)\}$. By our assumption, $\mathbb{P}(A^*) = 1-\alpha$. Using the formula for the portfolio:

$$\begin{aligned}
ES_\alpha(L_1 + L_2) &= \frac{1}{1-\alpha}\mathbb{E}[(L_1 + L_2)1_{A^*}] \\
&= \frac{1}{1-\alpha}\mathbb{E}[L_1 1_{A^*}] + \frac{1}{1-\alpha}\mathbb{E}[L_2 1_{A^*}]
\end{aligned}$$

Since $\mathbb{P}(A^*) = 1-\alpha$, we can apply our Lemma to L_1 and L_2 individually:

$$\mathbb{E}[L_1 1_{A^*}] \leq \mathbb{E}[L_1 1_{\{L_1 \geq VaR_\alpha(L_1)\}}] \quad \text{and} \quad \mathbb{E}[L_2 1_{A^*}] \leq \mathbb{E}[L_2 1_{\{L_2 \geq VaR_\alpha(L_2)\}}]$$

Substituting these bounds back into the equation:

$$\begin{aligned}
ES_\alpha(L_1 + L_2) &\leq \frac{1}{1-\alpha}\mathbb{E}[L_1 1_{\{L_1 \geq VaR_\alpha(L_1)\}}] + \frac{1}{1-\alpha}\mathbb{E}[L_2 1_{\{L_2 \geq VaR_\alpha(L_2)\}}] \\
&= ES_\alpha(L_1) + ES_\alpha(L_2)
\end{aligned}$$

Thus, Expected Shortfall is subadditive. Combining this with monotonicity, cash-invariance, and positive homogeneity, we conclude that ES_α satisfies all axioms of a coherent risk measure.

7 Mean-variance hedging

(1) Minimize $\mathbb{E}[L^2]$ over (V_0, ξ)

The hedging loss is given by $L = C - V_0 - \xi(S_T - S_0)$. We want to find the pair (V_0, ξ) that minimizes the expected squared loss, also known as the mean-squared error:

$$f(V_0, \xi) = \mathbb{E}[(C - V_0 - \xi(S_T - S_0))^2]$$

To find the minimum, we take the partial derivatives with respect to V_0 and ξ – and set them to zero. First, taking the derivative with respect to V_0:

$$\begin{aligned} \frac{\partial f}{\partial V_0} &= \mathbb{E}[-2(C - V_0 - \xi(S_T - S_0))] = 0 \\ \mathbb{E}[C] - V_0 - \xi\mathbb{E}[S_T - S_0] &= 0 \\ V_0 &= \mathbb{E}[C] - \xi\mathbb{E}[S_T - S_0] \end{aligned}$$

This gives us the formula for V_0.

Next, taking the derivative with respect to ξ:

$$\frac{\partial f}{\partial \xi} = \mathbb{E}[-2(S_T - S_0)(C - V_0 - \xi(S_T - S_0))] = 0$$

Substitute the expression we found for V_0 into this equation:

$$\begin{aligned} \mathbb{E}\left[(S_T - S_0)\left(C - (\mathbb{E}[C] - \xi\mathbb{E}[S_T - S_0]) - \xi(S_T - S_0)\right)\right] &= 0 \\ \mathbb{E}\left[(S_T - S_0)\left((C - \mathbb{E}[C]) - \xi(S_T - S_0 - \mathbb{E}[S_T - S_0])\right)\right] &= 0 \end{aligned}$$

Expanding the expectation:

$$\mathbb{E}[(S_T - S_0)(C - \mathbb{E}[C])] - \xi\mathbb{E}[(S_T - S_0)(S_T - S_0 - \mathbb{E}[S_T - S_0])] = 0$$

Recall the definition of covariance for two random variables X and Y: $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[X(Y - \mathbb{E}[Y])]$. Applying this to our terms:

- $\mathbb{E}[(S_T - S_0)(C - \mathbb{E}[C])] = \mathrm{Cov}(S_T - S_0, C)$
- $\mathbb{E}[(S_T - S_0)(S_T - S_0 - \mathbb{E}[S_T - S_0])] = \mathrm{Var}(S_T - S_0)$

Substituting these back into the expanded derivative equation:

$$\begin{aligned} \mathrm{Cov}(S_T - S_0, C) - \xi\mathrm{Var}(S_T - S_0) &= 0 \\ \xi &= \frac{\mathrm{Cov}(C, S_T - S_0)}{\mathrm{Var}(S_T - S_0)} \end{aligned}$$

(2) Interpret V_0 and ξ as a regression coefficient

The expressions we derived for V_0 and ξ perfectly match the ordinary least squares (OLS) estimators for a simple linear regression. If we were to regress the payoff C on the asset return $(S_T - S_0)$ using the model:

$$C = V_0 + \xi(S_T - S_0) + \epsilon$$

where ϵ is the error term (equivalent to our hedging loss L), the standard formulas for the slope and intercept that minimize the sum of squared errors are exactly the ξ and V_0 we just found. Thus, mean-variance hedging is mathematically equivalent to finding the best linear approximation of the derivative's payoff C using the underlying asset's price change $S_T - S_0$ & interpreting the residual as the unhedgeable risk.

(3) Binomial model exact hedging

In a one-period binomial model, the terminal stock price S_T can only take two possible values:

- Up state: $S_T^u = S_0(1+b)$
- Down state: $S_T^d = S_0(1+a)$

Similarly, the derivative payoff C will take two corresponding values, C_u and C_d.

We want to find (V_0, ξ) such that $L = 0$ almost surely. This means the portfolio must exactly match the payoff in both states:

$$V_0 + \xi(S_0(1+b) - S_0) = C_u \Rightarrow V_0 + \xi S_0 b = C_u \tag{1}$$
$$V_0 + \xi(S_0(1+a) - S_0) = C_d \Rightarrow V_0 + \xi S_0 a = C_d \tag{2}$$

This is a system of two linear equations with two unknowns (V_0, ξ). Because $a < 0 < b$, the determinant of the system's matrix is proportional to $(b-a)$, which is strictly positive (non-zero). Therefore, a unique solution exists. Because there are only two states of the world and two available assets (the risk-free asset and the stock), the market is complete, and we can replicate any payoff perfectly.

To find the solution, subtract the second equation from the first:

$$\begin{aligned}
(V_0 + \xi S_0 b) - (V_0 + \xi S_0 a) &= C_u - C_d \\
\xi S_0(b-a) &= C_u - C_d \\
\xi &= \frac{C_u - C_d}{S_0(b-a)}
\end{aligned}$$

This is the standard delta hedging ratio in the binomial model.

Now, substitute ξ back into the second equation to solve for V_0:

$$\begin{aligned}
V_0 &= C_d - \xi S_0 a \\
V_0 &= C_d - a\left(\frac{C_u - C_d}{b-a}\right) \\
V_0 &= \frac{C_d(b-a) - a(C_u - C_d)}{b-a} \\
V_0 &= \frac{bC_d - aC_d - aC_u + aC_d}{b-a} \\
V_0 &= \frac{bC_d - aC_u}{b-a}
\end{aligned}$$

This V_0 represents the initial replication cost (or arbitrage-free price) of the derivative in the binomial framework.

8 Hedging may increase risk

(1) Compute the optimal ξ

From Exercise 7, we know that the optimal hedging ratio that minimizes the mean-squared error is given by:

$$\xi = \frac{\mathrm{Cov}(C, S_T - S_0)}{\mathrm{Var}(S_T - S_0)}$$

We are given $S_T = S_0 + \epsilon$, with $\mathbb{E}[\epsilon] = 0$ and $\mathrm{Var}(\epsilon) = \sigma^2$. This implies $S_T - S_0 = \epsilon$, so the denominator is simply $\mathrm{Var}(\epsilon) = \sigma^2$.

The payoff is $C = S_T^2 = (S_0 + \epsilon)^2 = S_0^2 + 2S_0\epsilon + \epsilon^2$. Now we compute the covariance in the numerator:

$$\begin{aligned}\mathrm{Cov}(C, \epsilon) &= \mathrm{Cov}(S_0^2 + 2S_0\epsilon + \epsilon^2, \epsilon) \\ &= \mathrm{Cov}(S_0^2, \epsilon) + 2S_0\mathrm{Cov}(\epsilon, \epsilon) + \mathrm{Cov}(\epsilon^2, \epsilon)\end{aligned}$$

Since S_0^2 is a constant, its covariance with ϵ is 0. We know $\mathrm{Cov}(\epsilon, \epsilon) = \mathrm{Var}(\epsilon) = \sigma^2$. For $\mathrm{Cov}(\epsilon^2, \epsilon)$, we have:

$$\mathrm{Cov}(\epsilon^2, \epsilon) = \mathbb{E}[\epsilon^3] - \mathbb{E}[\epsilon^2]\mathbb{E}[\epsilon] = \mathbb{E}[\epsilon^3] - \sigma^2(0) = \mathbb{E}[\epsilon^3]$$

Assuming ϵ follows a symmetric distribution (such as a normal distribution), its skewness is zero, meaning $\mathbb{E}[\epsilon^3] = 0$. Thus, the covariance simplifies to:

$$\mathrm{Cov}(C, \epsilon) = 2S_0\sigma^2$$

Plugging this back into the formula for ξ:

$$\xi = \frac{2S_0\sigma^2}{\sigma^2} = 2S_0$$

(2) Compute the variance of the hedging error

The initial portfolio value is $V_0 = \mathbb{E}[C] - \xi\mathbb{E}[\epsilon]$. Since $\mathbb{E}[\epsilon] = 0$, $V_0 = \mathbb{E}[C]$.

$$\mathbb{E}[C] = \mathbb{E}[S_0^2 + 2S_0\epsilon + \epsilon^2] = S_0^2 + 2S_0(0) + \sigma^2 = S_0^2 + \sigma^2$$

The hedging error (loss) is $L_H = C - V_0 - \xi(S_T - S_0)$. Substituting our terms:

$$\begin{aligned}L_H &= (S_0^2 + 2S_0\epsilon + \epsilon^2) - (S_0^2 + \sigma^2) - 2S_0(\epsilon) \\ &= \epsilon^2 - \sigma^2\end{aligned}$$

The variance of the hedging error is $\mathrm{Var}(L_H) = \mathrm{Var}(\epsilon^2 - \sigma^2) = \mathrm{Var}(\epsilon^2)$. Assuming $\epsilon \sim \mathcal{N}(0, \sigma^2)$, the fourth moment is $\mathbb{E}[\epsilon^4] = 3\sigma^4$.

$$\begin{aligned}\mathrm{Var}(\epsilon^2) &= \mathbb{E}[\epsilon^4] - (\mathbb{E}[\epsilon^2])^2 \\ &= 3\sigma^4 - (\sigma^2)^2 \\ &= 2\sigma^4\end{aligned}$$

So, the variance of the hedging error is $2\sigma^4$.

(3) Hedging reduces risk for small σ, but may increase tail losses for large σ

Let us define the unhedged loss L_U, which occurs if we simply price the derivative at its expected value V_0 but hold no stock ($\xi = 0$):

$$\begin{aligned}L_U = C - V_0 &= (S_0^2 + 2S_0\epsilon + \epsilon^2) - (S_0^2 + \sigma^2) \\ &= 2S_0\epsilon + \epsilon^2 - \sigma^2\end{aligned}$$

We can compare the variance (risk) of both strategies:

$$\mathrm{Var}(L_U) = \mathrm{Var}(2S_0\epsilon + \epsilon^2) = 4S_0^2\sigma^2 + 2\sigma^4$$

Since $\mathrm{Var}(L_H) = 2\sigma^4 < 4S_0^2\sigma^2 + 2\sigma^4$, hedging strictly reduces the overall variance. For small σ, the $4S_0^2\sigma^2$ term heavily dominates the unhedged variance, meaning the linear hedge $\xi = 2S_0$ is highly effective at stabilizing the portfolio.

However, let us examine the extreme tail events (large values of $|\epsilon|$), which become more probable for large σ. Suppose the market crashes, resulting in a large negative shock $\epsilon < -2S_0$. In this scenario, the linear term $2S_0\epsilon$ becomes a large negative number.

$$L_U = \epsilon^2 + 2S_0\epsilon - \sigma^2 \quad \text{while} \quad L_H = \epsilon^2 - \sigma^2$$

Because $2S_0\epsilon < 0$, it strictly follows that:

$$L_U < L_H$$

Conclusion: During a severe market downturn, the unhedged portfolio actually loses less money than the hedged portfolio. The hedge requires holding a long position of $\xi = 2S_0$ shares. If the stock price plummets, this long position incurs massive losses. The quadratic payoff $C = S_T^2$ stops dropping once it hits zero (it cannot be negative), meaning the derivative no longer offsets the linear losses from the stock hedge. Therefore, for large σ where extreme market movements are possible, the mean-variance hedge introduces dangerous tail risk.